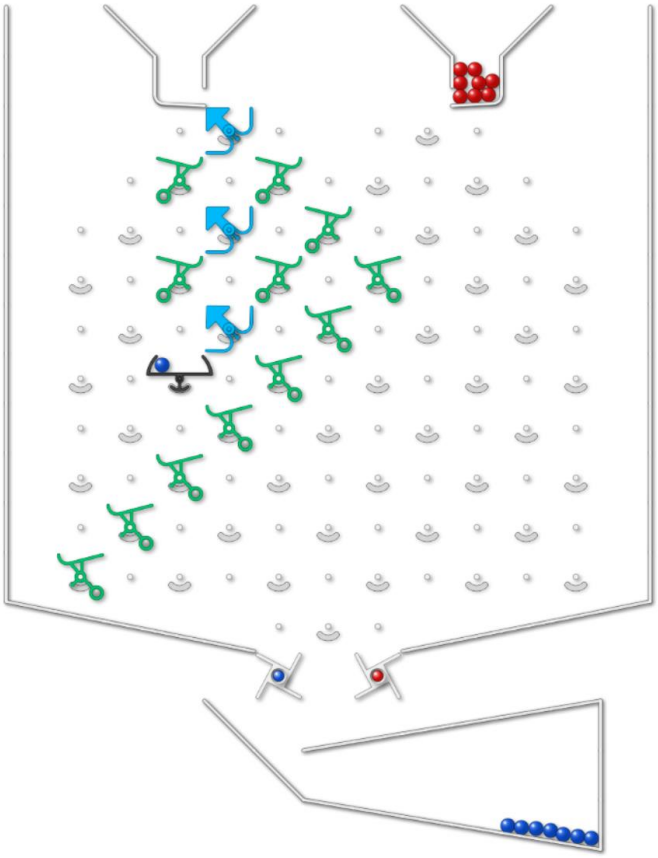
Week 1 Karan Chand s1033357

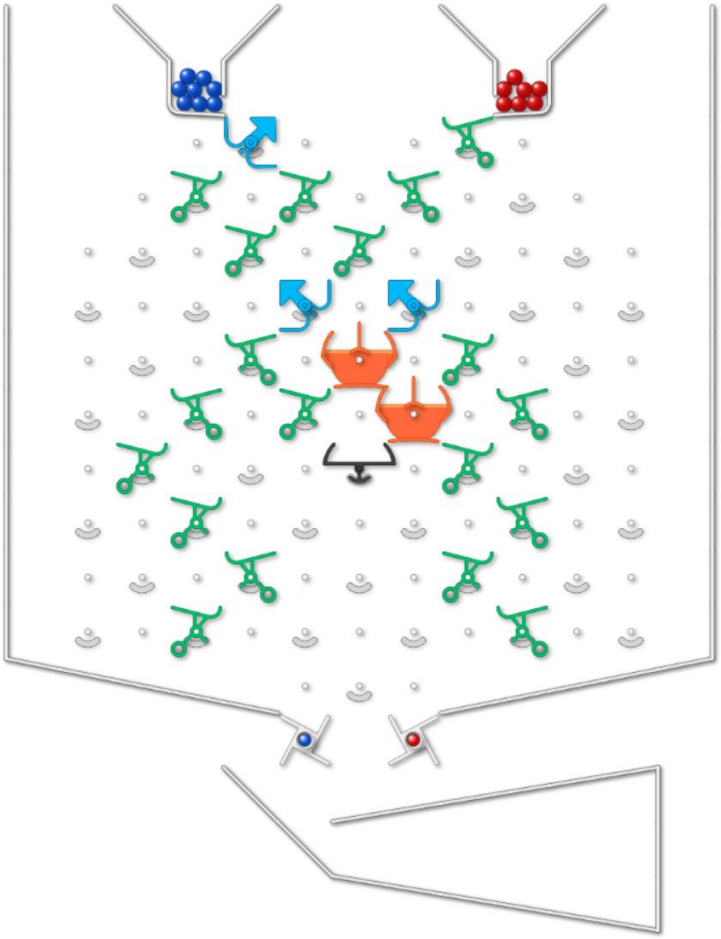
Exercise 1.1



Explanation

The red marbles are never supposed to drop, so the blue marbles stay on the left side. In the starting position, the marbles are all pointing left, just like the end position. The goal is setting up the bits for the last marble. With this configuration, 7 marbles tumble through the system, while the last marble hits the last bit, causing it to turn left. The other marbles all went to the right of that bit.

Exercise 1.2



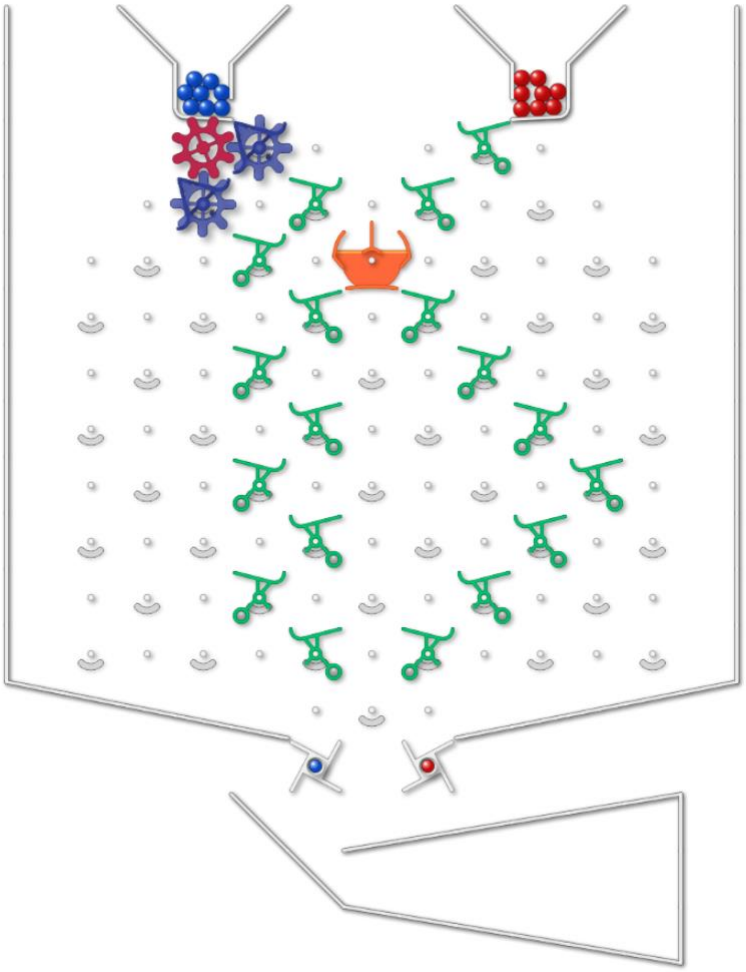
Explanation

Truth table:

A	B	OUT	Explanation
0	0	0	A and B 0 -> point left, 1 -> point right
1	0	0	OUT 0 -> red, 1 -> blue
0	1	0	
1	1	1	

The highest bit that was added is for checking both of the blue marble conditions. This starts pointing to the right so the bit A is checked first. So, this bit is only used twice when bit A is 1. Else it gives a red marble. This red marble is accepted regardless of bit B. If bit A is 0, the red marble is called and accepted. If both bits are 1, the blue marble is called again and accepted.

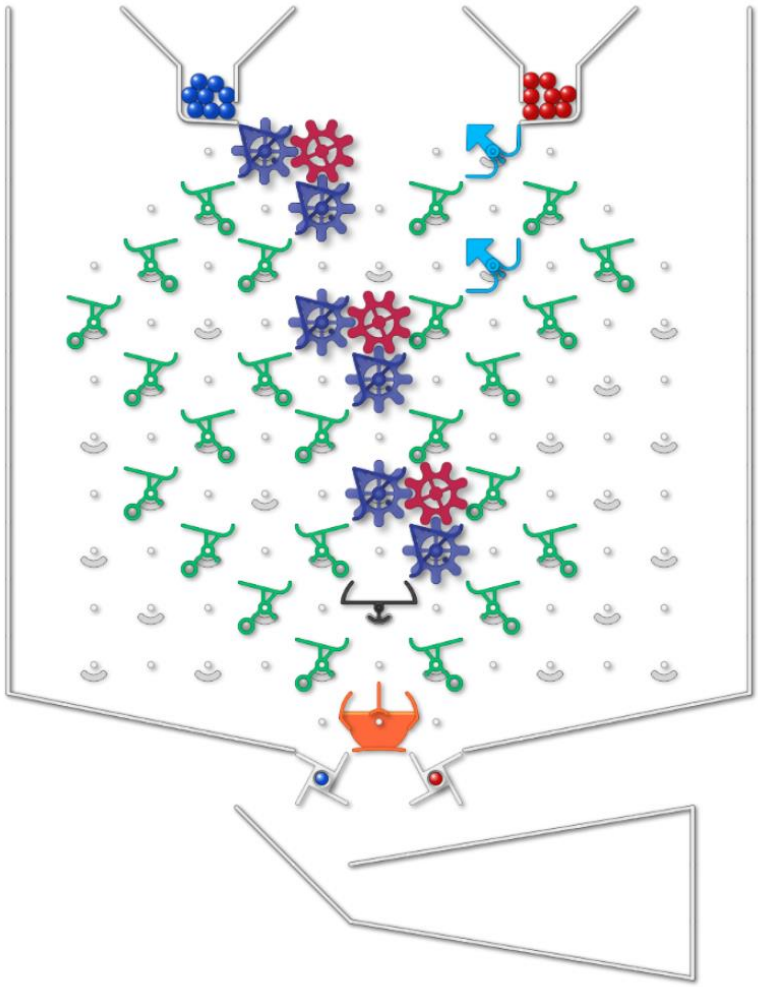
Exercise 1.3



Explanation

First the switch is pointing left, which means that the first marble is going right. After the first blue marble has gone to the right, the switch permanently directs the marbles to the left side. Now a red marble is dropped. This is lead to the left side, dropping another blue marble. All the blue marbles are now being dropped sequentially, as the switch resets itself.

Exercise 1.3



Explanation

The gear bit groups basically represent switches as seen in the previous exercise. If they are turned to the left, they don't act like a switch, but let the marbles through to the next group of gear bits. This movement represents that a marble should not be counted. If the group is pointed to the right though, a red marble is called, the bits keep track of the amount of marbles passed in binary and another blue marble is called. This keeps going until all the groups are pointing to the left, which means that counting is done and only 1 blue marble is going to be let through to the collection pit.

How the bits keep track of red marbles: The highest bit represents the right bit and the lowest bit the left bit. Counting is done by first turning the right one. When the next marble flows through, it passes both bits, turning them both and counting up. When the next marble is passed through, it resets again, passing only the right bit to count to 3. When red marbles pass through the system, a blue marble is called and when a blue marble passes through the system, a red marble is called or the run is finished.